Tianyang **Feng**

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Education

Nanyang Technological University — *Singapore*
M.Sc. in Artificial Intelligence — *Jan. 2026 - Present*

Tongji University — *Shanghai, China*
B.Eng. in Optoelectronic Information Science and Engineering — *Sep. 2020 - Jun. 2024*

Publications (* equal contribution)

Edge Accelerated Reconstruction Using Sensitivity Analysis for Single-lens Computational Imaging.
Xuquan Wang*, Tianyang Feng*, Yujie Xing*, Ziyu Zhao, Xiong Dun, Zhanshan Wang, Xinbin Cheng
Advanced Imaging, 2025, 2(3): 031001 (Editors' Pick) [Paper]

Experience

Tongji University — *Shanghai, China*
Research Assistant at IPOE, advised by Prof. Xiong Dun — *Aug. 2024 - Jul. 2025*

- Focusing on the topic of *Edge Accelerated Reconstruction Using Sensitivity Analysis for Single-lens Computational Imaging* as the team leader, in which we propose a new model compression strategy with better performance in reconstruction quality and speed.
- Responsible for all the research cycle, including model design, training, development of compression algorithms, performance verification, and co-authoring the paper with the supervisor.
- Proved that among the algorithms currently capable of being deployed at the edge, the simultaneous application of pruning and quantization achieves a balance between high restoration metrics (PSNR, SSIM) and speed metrics (FPS) in the field of image reconstruction.

Projects

Safety Evaluation Framework for Generative Autonomous Driving Video Models — *Singapore*
Designer & Developer, Team Member of 3; Top 10% Course Project. — *Jan. 2026 - Apr. 2026*

- Built a safety evaluation system to identify semantic inconsistencies, temporal logic violations, and unsafe driving decisions in adversarially attacked generative autonomous-driving videos.
- Co-proposed a structured multi-judge VLM framework with confidence-aware aggregation, co-designed the evaluation metrics, implemented video serialization, and contributed to result analysis.
- Benchmarked GPT-4.1-mini and Qwen2.5-VL-32B-Instruct on 100 adversarial driving videos; achieved 0.792 balanced accuracy, earning Top 10% course-project recognition.

Full-process Development & Deployment of Infrared Computational Imaging — *Shanghai, China*
Designer & Developer, Team Member of 4. — *Aug. 2024 - Feb. 2025*

- Developed an infrared computational imaging camera that utilizes a self-designed meta-surface lens in conjunction with existing sensors and chips. Leveraged computational power to replace complex optical structures to build a lightweight imaging system.
- Focused on the design, training, compression, and deployment of algorithms, as well as the development of imaging, display, recognition, and other applications on edge devices (Jetson, RK3588).
- This infrared imaging system is lighter and provides clearer images than traditional optical systems. It integrates YOLO into the network for target recognition, making it suitable for various aircraft.

Battle Ground – A Real-time Shooting Game — *Shanghai, China*
Designer & Developer, Team Member of 2. [CODE] — *Apr. 2021 - Jun. 2021*

- Built a C++ real-time shooting game with scene modules for the main and settings interfaces, character modules for players, enemies, and firearms, and detection for ammunition, health, collisions, and score rankings.
- Implemented background music, a timing system, character movement, reloading, and weapon switching.
- Enabled mouse-controlled aiming and shooting; generated randomized ammunition drops and collision-based ammunition pickup.

Skills

Programming	Python, C/C++, MATLAB, LaTeX
Frameworks & Tools	Pytorch, Linux, Git, Vim, Docker
Languages	Chinese (native), English (fluent)